Mattan S. Ben-Shachar, PhD

As a freelance research analyst and statistical consultant specializing in the social sciences, I offer comprehensive guidance throughout all stages of the research process, having provided support and consultation in these areas to numerous researchers, across several institutions. Holding a PhD in neurocognitive psychology from Ben-Gurion University, I am deeply rooted in theoretical and applied statistics, and am committed to the principles of data analysis, inferential statistics, and reproducible research.

Concurrently, I am a statistics educator at both Ben-Gurion University and Tel-Aviv University, teaching bachelor- and graduate-level statistics, research methods, and R programming, ranging from introductory statistics to applied machine learning. I supplement my goal of making statistics more accessible as an active `R` developer, creating tools that aid in understanding statistical models.

Contact Info

mattansb@msbstats.info
Home Page | Blog BlueSky | LinkedIn GitHub ORCID | G-Scholar

Education

2023
Ph.D. in Psychology
Ben-Gurion University — Be'er Sheva, Israel

Title: *The Electrophysiological Basis of Processing Speed* (Download dissertation (pdf) | Audio overview) Supervisor: Prof. Andrea Berger

2017
M.A. in Experimental Cognitive Psychology
Ben-Gurion University — Be'er Sheva, Israel

Title: *Electrophysiological Evidence for the Role of Processing Speed and Attentional Control in Visual Recognition Tasks.* Supervisor: Prof. Andrea Berger

2015
B.A. in Behavioral Sciences
Ben-Gurion University — Be'er Sheva, Israel

Graduated *Magna Cum Laude*

Professional Experience

Since 2016
Statistical Consultant & Research Analyst

I am deeply rooted in theoretical and applied statistics, and am committed to the principles of data analysis, inferential statistics, and reproducible research, while also mindful of the practical needs of researchers and academics. I provide guidance throughout all stages of the research process and decision-guided statistical analyses. I specialize in the following methods:

- Generalized linear models (regression, AN(C)OVA)
- Generalized mixed effects models (HLM, LMM)
- Missing data analysis and treatment
- Structural equation modeling (SEM)
- Factor analysis and clustering
- Bayesian modeling and inference
- Nonlinear models (NLM)

For more info and client testimonials, see my home page.

Languages

English - native language
Hebrew - native language

Technical Skills

Programming: R, Matlab, SQL, Stan, Python, Git(Hub) (See my github profile).

Documentation: Office Suites, (R)Markdown/Quatro, Google Docs/Sheets.

See short resume.

Last updated on 2026-01-11.

Since 2015

Statistics Lecturer

My teaching and instructing are focused on undergraduate- and graduate-level statistics, research methods, and R programming. Over the last 10 years I have taught over 1,560 undergraduate students and over 930 graduate students across over a dozen courses. See *Teaching Experience* section.

Since 2016

R Developer

Core developer of the *easystats* eco-system of packages – an award-winning collection of R packages, which aims to provide a unifying and consistent framework to tame, discipline and harness the scary R statistics and their pesky models (e.g., *parameters*, *performance*, and more).

- Developer and maintainer of *effectsize* for estimating indices of effect size and standardized parameters.
- Developer of *bayestestR*, a package for describing and testing posterior distributions and Bayesian models.

See my full github profile.

List of additional packages I have contributed to: `afex`, `brms`, `ggeffects`, `semTools`.

2015 | 2022

Neurocognitive Researcher

My research focused on differential cognitive psychology using behavioral and physiological methods, occasionally collaborating with international research groups. I have experience designing computerized tasks that evoke individual differences in an array of cognitive abilities. I have also led and managed research projects, including teams of graduate students and research assistants.

Neurophysiological Signal Processing: I have collected and analyzed multi-channel electroencephalography (EEG) data using `eeglab`. My experience with signal pre-processing includes noise "cleaning" methods as well as dimension reduction tools (PCA, ICA); I specialize in latency measurement of event-related potentials (ERPs).

- Developed Matlab toolboxes for artifact removal (TBT, an eeglab plugin) and for plotting and extracting measurements for processed EEG/ERP/wavelet data (EPP-TB).
- Instructed workshops for pre-processing of experimental EEG.

Since 2020

MITAM Exam Developer

National Institute for Testing and Evaluation

I am a developer and reviewer of the MITAM exam (entrance test for advanced degrees in psychology - equivalent to the GRE).

Teaching Experience

Since 2018

Lecturer & Teaching Fellow

School of Psychological Sciences, Tel-Aviv University

Graduate level courses: - Data Science Lab for Psychologists (since 2023) - Hierarchical Models (since 2023)

Department of Psychology, Ben-Gurion University

Graduate level courses: - Advanced Research Methods for Psychologists (since 2026) - Machine Learning for Psychologists (since 2022) - Essential Data Science Skills (since 2022) - Hierarchical Linear Models (since 2022, biannually) - Structural equation modelling (2026)

Under-graduate level courses: - Regression and Multi-Variables Analysis (2022–2023) - Experimental Psychology (2018)

Statistics and Data Program, Ben-Gurion University of the Negev

Under-graduate level courses: - Bayesian Statistics and Modeling (since 2023) - Introduction to Data Science (2023–2025)

School of Cognitive Sciences, Ben-Gurion University of the Negev

Under-graduate level courses: - Inferential Statistics for Cognitive Sciences (2022)

Workshops

- *Advance R Tools for Research.* (2 days; Sep 2025)
- *Bayesian Statistics: So Much More Than Just the Bayes factor.* (2 days; Aug 2024; Mar 2025)
- *Evaluating Evidence and Making Decisions using Bayesian Statistics*. Workshop at the The 8th Conference on Cognitive Research of the Israeli Society for Cognitive Psychology, Israel.
- *Intro to the R Programming Language* (7h; Jan 2021, Feb 2023)
- *Trends in Open Science* (2h; Dec 2018, May 2019, Nov 2019)
- *ANOVA and Contrasts in R* (2h; May 2018)

2015 | 2018

Teaching Assistant & Instructor

Ben-Gurion University

Department of Psychology, Ben-Gurion University

Graduate level courses: - Practical applications in R (2019–2021) - Structural equation modelling in R (2020–2021) - Analysis of factorial designs in R (2019)

Under-graduate level courses: - Experimental Psychology (Fall + Summer 2017) - Inferential Statistics (2017–2018) - Introduction to Statistics (2015)

Department of Health Systems Management

Graduate level courses: - Research Seminar (*for Graduate Students*) (2016–2017)

Awards And Scholarships

* Denotes upcoming courses.

2023 **SIPS Mission Award**
Society for Improving Psychological Science (SIPS)
Awarded to the `easystats` project for improving the training and research practices of psychological scientists.

2015 | 2022 **Full academic scholarship for graduate studies**
Humanities and Social Sci Faculty, Ben-Gurion University
Be’er Sheva, Israel

2020 | 2021 **Negev Mid-Way Scholarship**
Kreitman School of Advanced Graduate Studies — Be’er Sheva, Israel

2020 **SIPS Commendation**
Society for Improving Psychological Science (SIPS)
Awarded to the `bayestestR` package for improving the training and research practices of psychological scientists.

2020 **M.H.R Publication Award**
Humanities and Social Sci Faculty, Ben-Gurion University
Be’er Sheva, Israel
For *Ben-Shachar et al. (2020). DOI:10.1111/acer.14244*

2018 **Erasmus+ EU Travel Grant**
EU programme for education, training, youth and sport in Europe

2018 **Radboud Summer School Scholarship**
Radboud University — Nijmegen, Netherlands

Studied Linear Algebra for Neuroscience with Prof. Mike X. Cohen

2014 **Award for Academic Distinction**
Dep. of Sociology and Anthropology, Ben-Gurion University
Be’er Sheva, Israel

Selected Publications

Peer Reviewed Publications

- Hochman, S.*, Ben-Shachar, M.S.*, Kadosh, R.C., & Henik, A. (2026). A novel task for measuring numerical bias among adults. *Cognition, 271*, 106439. DOI:10.1016/j.cognition.2026.106439
- Ben-Shachar, M.S., & Berger, A. (2024). Visual-Spatial Abilities Are NOT Related to the Speed of Mental Rotation. *Journal of Experimental Psychology: Human Perception and Performance.* DOI:10.31234/osf.io/gkzxf
- Thériault, R., Ben-Shachar, M.S., Patil, I., Lüdecke, D., Wiernik, B. M., & Makowski, D. (2024). Check your outliers! An introduction to identifying statistical outliers in R with easystats. DOI:10.31234/osf.io/bu6nt
- Ben-Shachar, M.S., Patil, I., Thériault, R., Wiernik, B.M., Lüdecke, D. (2023). Phi, Fei, Fo, Fum: Effect Sizes for Categorical Data That Use the Chi-Squared Statistic. *Mathematics, 11*, 1982. DOI:10.3390/math11091982
- Ben-Shachar, M.S., Lüdecke, D., & Makowski, D. (2020). effectsize: Estimation of Effect Size Indices and Standardized Parameters Aims of the Package. *Journal of Open Source Software, 5*(56), 2815. DOI:10.21105/joss.02815
- Ben-Shachar, M.S., Shmueli M., Meintjes, E.M., Molteno, C.D., Jacobson, J.L., Jacobson, S.W., & Berger A. (2020). Prenatal alcohol exposure alters error detection during simple arithmetic processing: An Electroencephalography study. *Alcoholism: Clinical and Experimental Research, 44*(1), 114-124. DOI:10.1111/acer.14244
- Makowski, D., Ben-Shachar, M.S., Chen, S.A., & Lüdecke, D. (2019). Indices of Effect Existence and Significance in the Bayesian Framework. *Frontiers in Psychology, 10*, 2767. DOI:10.3389/fpsyg.2019.02767

Books and Book Chapters

- Jané, M., Xiao, Q., Yeung, S., Ben-Shachar, M.S., … & Feldman, G. (2024). *Guide to Effect Sizes and Confidence Intervals*. https://matthewbjane.quarto.pub/
- Ben-Shachar, M.S., & Berger, A. (2018). An Introduction to Attention and Its Implication for Numerical Cognition. In A. Henik & W. Fias (Eds.), *Heterogeneity of Function in Numerical Cognition* (pp. 93-110). Academic Press. DOI:10.1016/B978-0-12-811529-9.00005-4

Misc.

- Ben-Shachar, M.S.*, Lisson, S.*, Shotts-Peretz, D., Hannula-Sormunen, M., & Berger, A. (*preprint*). Spontaneous Focusing on Numerosity Linked to Quantity Discrimination in Children and Adults. DOI:10.31234/osf.io/mt3n9

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For a full list of publications, see my ORCID or Google Scholar profiles. Contact me a PDF of any of my publications.

In many of these works I was involved in the capacity of a statistical consultant or research analyst.

** Denotes equal contribution.*